Fixed-Time Angle Tracking Control for Multi-DOF Manipulator Driven by Pneumatic Artificial Muscles

Xin Liu, Xiaochen Zhang, Fang Xu, Shaomeng Gu, and Jinhui Zhang

***Abstract*—In this article, a fixed-time angle tracking control strategy is proposed to improve control performances of the multi-DOF manipulator driven by pneumatic artificial muscles (PAMs). The proposed fixed-time angle tracking control strategy includes a fixed-time disturbance observer and a fixed-time controller to estimate and compensate unknown disturbance in the multi-DOF manipulator, respectively. The fixed-time convergence of both the fixed-time disturbance observer and the multi-DOF manipulator with the fixed-time controller is proved by using homogeneous theory and Lyapunov strategy, respectively. Finally, experimental results show that the desired angle tracking control performances of the multi-DOF manipulator are achieved.**

***Index Terms*—Fixed-time controller, fixed-time disturbance observer, multi-DOF manipulator, pneumatic artificial muscles (PAMs).**

I. Introduction

In recent years, pneumatic artificial muscles (PAMs) attract more and more attention owing to the advantages of natural compliance, high power-weight ratio, safety, low cost, etc. [1], [2], [3]. They are widely used in various fields, such as pneumatic motion platforms [4], soft continuum manipulators [5], humanoid arm robots [6], and ankle-foot orthosis [7]. It should be noticed that, the control performances of systems driven by PAMs are difficult to reach a satisfactory level due to unknown disturbance, which is caused by compressibility of air, loss of piping pressure, hysteresis, nonlinearity of materials, unmodeled dynamics, etc. [8]. To deal with those issues, some control strategies are proposed to improve control performances of the systems driven by PAMs, for example, a super-twisting control strategy is designed to deal with uncertainty and external disturbance of joint systems driven by PAMs [9]. In [10], a sliding mode control strategy based on exponential reaching law is proposed to obtain good tracking performances of the systems driven by PAMs. An adaptive proxy-based robust control strategy is developed to handle the unknown disturbance of the systems driven by PAMs [11]. In [12], a reinforcement learning-based robust motion control strategy is proposed to achieve satisfactory and efficient control performances of robotic arm systems driven by PAMs. However, the convergence time of most of the existing control strategies for the multi-DOF manipulator driven by PAMs is infinite or decided by initial conditions. Recently, fixed-time control strategies are developed because the convergence time is settled and unrelated to the initial conditions. Therefore, it is a meaningful research to achieve the fixed-time angle tracking control for the multi-DOF manipulator driven by PAMs, which motivates us to make an in-depth study in this article.

It should be noted that, in the practical multi-DOF manipulator driven by PAMs, the control performances are always damaged by the unknown disturbance. To deal with the unknown disturbance, many control strategies based on disturbance observers are proposed to apply in various systems, such as stable flight control of unmanned aerial vehicles [13], velocity tracking control of gimbal servo systems [14], and current control of permanent-magnet synchronous motors [15]. In [16], a prescribed-time control strategy based on the disturbance observers is proposed to deal with the unknown disturbance of vehicles. A finite-time composite antidisturbance control strategy based on the asynchronous disturbance observers is developed to estimate and compensate the unknown disturbance of nonhomogeneous Markovian jump systems [17]. In [18], a model-free trajectory tracking control strategy based on the data-driven adaptive disturbance observer is proposed to solve the total disturbances of maritime autonomous surface ships. To deal with the unmodeled dynamics and unknown disturbances, an adaptive finite-time tracking control method is designed for a class of strict-feedback nonlinear systems [19]. At present,

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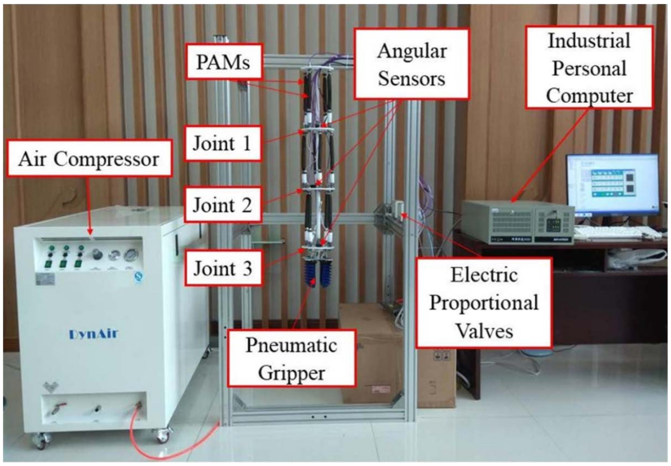


Fig. 1. Experiment platform of the multi-DOF manipulator.

most of the existing disturbance observers can only guarantee asymptotic convergence or finite-time convergence. Compared with finite-time control method, the fixed-time control method shows obvious advantages in convergence time, which is independent of the initial conditions. During the system control process, if the initial conditions are not accurate or even difficult to obtain, the convergence time of the system may be inaccurate or impossible to calculate, which may affect the system control performance. The fixed-time disturbance observers are widely used in various systems, such as space manipulator systems [20], underactuated surface vehicles [21], reusable launch vehicle [22], and multi agent systems [23].

In this study, a fixed-time angle tracking control strategy is proposed to achieve desired angle tracking control performances of the multi-DOF manipulator driven by PAMs. To estimate and compensate the unknown disturbance of the multi-degree-of-freedom (DOF) manipulator, the fixed-time disturbance observer and the fixed-time controller are designed, respectively. Moreover, the fixed-time convergence of the proposed angle tracking control strategy is analyzed by using homogeneous theory and Lyapunov strategy. Finally, angle tracking experimental results of different amplitudes, frequencies, loads and initial angle tracking errors demonstrate that the proposed fixed-time angle tracking control strategy is effective to achieve desired angle tracking control performances for the multi-DOF manipulator.

Notations: In this article, the superscript "T" means the transpose of a matrix. For any $a \geq 0$, $\lceil \chi \rfloor^a$ is defined as $\lceil \chi \rfloor^a = |\chi|^a \text{sign}(\chi)$, $\chi \in \mathbb{R}$, and $(d\lceil \chi \rfloor^a / d\chi) = a|\chi|^{a-1}$. $\Upsilon(a_1, a_2) = a_1/(1+a_1)(1+a_2)$ for any $a_1 \geq 0$ and $a_2 \geq 0$. $|\cdot|$ and $\|\cdot\|$ stands for the absolute value and Euclidean norm, respectively.

II. Problem Formulation

A. System Structure

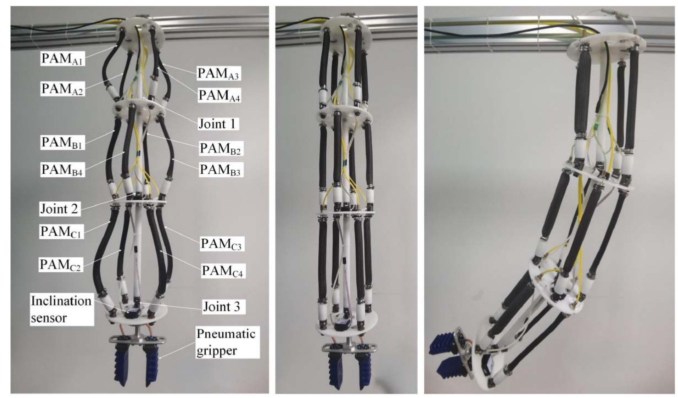
The experiment platform of the multi-DOF manipulator driven by PAMs is shown in Fig. 1. In Fig. 1, the multi-DOF manipulator consists of the control system, driving system and measurement system. The control system includes an industrial personal computer (IPC), a D/A card and thirteen electric proportional valves. The driving system of consists of a pneumatic gripper, an air compressor, and twelve PAMs. The measurement system includes three angular sensors. The model number and manufacturer of components for the pneumatic soft bending actuator control system is shown in Table I.

The working principle of the PAM relies on the inflation and deflation of the rubber tube encased in a braided mesh shell. Inject compressed air into the PAM, which contracts radially and outputs tensile force axially. The maximum contraction and maximum allowable pressure of the PAMs in this article are 22% and 0.7 MPa, respectively. In the control process of multi-DOF manipulator, angles are obtained by the three angular sensors, then the angles are transmitted to the IPC. The pressures are transmitted to the thirteen electric proportional valves by the D/A card. Then the thirteen electric proportional valves are used to regulate internal pressures of the PAMs and the pneumatic gripper. Compressed air is injected into the PAM which contracts radially and outputs tension axially. The compressed air is generated as power source by the air compressor. Thirteen electric proportional valves are used to adjust internal pressures of twelve PAMs and the pneumatic gripper. Pulling forces for six groups of antagonistic PAMs are produced by compressed air charging and discharging through twelve electric proportional valves.

The control states of the multi-DOF manipulator include: natural state, preloading state, and movement state, as shown in Fig. 2. The natural state is that all the PAMs are without

TABLE I
Model Number and Manufacturer of Components for the Multi-DOF Manipulator Control System

Component	Model Number	Manufacturer
IPC	610H	ADVANTECH
D/A card	PCI-1724U	ADVANTECH
Electric proportional valve	ITV0050	SMC
Pneumatic gripper	FMA2-M5086	SRT
Air compressor	DA7003CS	DMT
Angular sensor	WT901C485	WITMOTION



(a) (b) (c)

Fig. 2. Control states of the multi-DOF manipulator. (a) Natural state. (b) Preloading state. (c) Movement state.

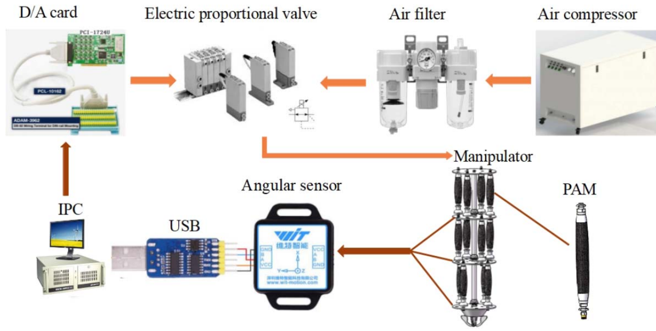


Fig. 3. Control process of the multi-DOF manipulator.

compressed air, as shown in Fig. 2(a). The preloading state is shown in Fig. 2(b): the preloaded internal pressures of compressed air for all PAMs are set at 0.2 MPa, then preloaded pulling forces are generated by the preloaded internal pressures for the six groups of antagonistic PAMs. The movement state is shown in Fig. 2(c): one group of PAMs ($\mathrm{PAM_{A1}}$, $\mathrm{PAM_{A2}}$, $\mathrm{PAM_{B1}}$, $\mathrm{PAM_{B4}}$, $\mathrm{PAM_{C1}}$, $\mathrm{PAM_{C2}}$) increases pressure of the compressed air and the other group ($\mathrm{PAM_{A3}}$, $\mathrm{PAM_{A4}}$, $\mathrm{PAM_{B2}}$, $\mathrm{PAM_{B3}}$, $\mathrm{PAM_{C3}}$, $\mathrm{PAM_{C4}}$) decreases same pressure by adjusting control signals of electric proportional valves.

The control process of the multi-DOF manipulator is shown in Fig. 3. In Fig. 3, the IPC sends the control signals to the electric proportional valves by the D/A card, internal pressures of the PAMs are regulated in real time by the electric proportional valves, where the compressed air is purified by the air filter. PAMs output tension to drive joint rotation, then the deflection motion of the multi-DOF manipulator is achieved. Simultaneously, the angle signals are transmitted to the IPC by the USB module.

B. Dynamical Model

According to [24], the dynamical model of the multi-DOF manipulator is expressed as

$$\mathcal{H}(\theta)\ddot{\theta} + \mathcal{C}(\theta,\dot{\theta})\dot{\theta} + \mathcal{G}(\theta) = \mathcal{M} + \mathcal{M}_d \tag{1}$$

where $\mathcal{H}(\theta) \in \mathbb{R}^{3\times 3}$ is the inertia vector, $\mathcal{C}(\theta,\dot{\theta}) \in \mathbb{R}^{3\times 3}$ stands for the centrifugal and coriolis vector, $\mathcal{G}(\theta) \in \mathbb{R}^{3\times 1}$ represents gravitational force vector, $\mathcal{M} = \mathcal{B}U \in \mathbb{R}^{3\times 1}$ indicates the control torque vector, $\mathcal{B} \in \mathbb{R}^{3\times 3}$ is the parameter matrix of the multi-DOF manipulator, $U \in \mathbb{R}^{3\times 1}$ is the control vector, $\mathcal{M}_d \in \mathbb{R}^{3\times 1}$ denotes the external disturbance, θ is the output angle vector. Let $X_1 = \theta$, $X_2 = \dot{\theta}$, then the dynamical model of the multi-DOF manipulator (1) is described as

$$\dot{X}_1 = X_2 \tag{2}$$
$$\dot{X}_2 = \mathcal{B}_0 U + D \tag{3}$$

where $D = \mathcal{H}(\theta)^{-1}(\mathcal{M}_d - \mathcal{C}(\theta,\dot{\theta})\dot{\theta} - \mathcal{G}(\theta))$, $\mathcal{B}_0 = \mathcal{H}(\theta)^{-1}\mathcal{B}$. Note that the term D is regarded as the unknown disturbance including compressibility of air, nonlinearity of PAMs, uncertainty of the model and external disturbance.

Subsequently, some useful assumptions, definitions and lemmas are introduced.

Consider the nonlinear system

$$\dot{\chi}(t) = g(\chi(t)), \chi(0) = \chi_0 \tag{4}$$

where $\chi = [\chi_1,\ldots,\chi_n]^T \in \mathbb{R}^n$ is the state vector, $g(\chi)$: $\mathbb{R}^n \to \mathbb{R}^n$ is the vector field. Assume that the origin is an equilibrium point of the nonlinear system (4). For the weight vector $d = [d_1,\ldots,d_n] \in \mathbb{R}^n$, $d_i > 0$ with $i = 1,\ldots,n$, the dilation mapping is defined as $\Pi_c^d = [c^{d_1}\chi_1,\ldots,c^{d_n}\chi_n]^T$, where $c > 0$.

Definition 1 [25]: The origin of the nonlinear system (4) is said to be globally finite-time stable if it is Lyapunov stable and for any $W > 0$ there exists $T > 0$, such that any trajectory starting within the ball $\|\chi\| \le W$ stabilizes at the origin in time T. Moreover, if there exists a fixed constant $T_{\max} > 0$ such that $T \le T_{\max}$ for any χ_0, the origin of the nonlinear system (4) is said to be fixed-time stable.

Definition 2 [26]: If $g(\Pi_d(\chi)) = c^k \Pi_c^d g(\chi)$ for any $c > 0$ and $\chi \in \mathbb{R}$ holds, then the vector field g is said to be d-homogeneous of degree k.

Definition 3 [27]: A vector field is homogeneous in the bi-limit if it is homogeneous in both the 0-limit and the ∞-limit.

Lemma 1 [28]: For the nonlinear system (4), suppose that g is the homogeneous vector field in the bi-limit with associated triples (d_0, λ_0, g_0) and $(d_\infty, \lambda_\infty, g_\infty)$. If the origins of the nonlinear systems $\dot{\chi} = g(\chi)$, $\dot{\chi} = g_0(\chi)$ and $\dot{\chi} = g_\infty(\chi)$ are globally asymptotically stable, then the following statements hold.

1) The origin of the nonlinear system (4) is fixed-time stable if $\lambda_0 < 0 < \lambda_\infty$.
2) The real numbers λ_{V_0} and λ_{V_∞} satisfy $\lambda_{V_0} > \max d_{0i}$ and $\lambda_{V_\infty} > \max d_{0i}$, $i = 1,\ldots,n$. There exists a continuous, positive definite and proper function $V(\chi)$ such that the function $\partial V/\partial \chi_i$ is homogeneous in the bi-limit with triples $(d_0, \lambda_{V_0} - d_{0i}, \partial V_0/\partial \chi_i)$ and $(d_\infty, \lambda_{V_\infty} - d_{\infty i}, \partial V_\infty/\partial \chi_i)$, and $(\partial V/\partial \chi_i) g(\chi) < 0$.

Lemma 2 [27]: Consider the perturbed system $\dot{\chi} = g(\chi, \Delta)$, where $\Delta = [\Delta_1,\ldots,\Delta_n]^T \in \mathbb{R}^n$ is the unknown disturbance. The vector field g is homogeneous in the bi-limit with associated triples $((d_0,\delta_0),\lambda_0,g_0)$ and $((d_\infty,\delta_\infty),\lambda_\infty,g_\infty)$, where δ_0 and δ_∞ are the weights associated with the unknown disturbance Δ. If the origin of the nominal system $\dot{\chi} = g(\chi,0)$ is fixed-time stable, then for all $\Delta \in \mathbb{R}^n$, the function V given by Lemma 1 satisfies

$$\frac{\partial V}{\partial \chi} g(\chi,\Delta) \le -a_V \Upsilon\left(V^{\frac{\lambda_{V_0}+\lambda_0}{\lambda_{V_0}}}, V^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\lambda_{V_\infty}}}\right) + a_\Delta \sum_{i=1}^{n} \Upsilon\left(|\Delta_i|^{\frac{\lambda_{V_0}+\lambda_0}{\delta_{0,i}}}, |\Delta_i|^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\delta_{\infty,i}}}\right)$$

where $a_V > 0$, $a_\Delta > 0$.

Assumption 1 [29]: The derivative of the unknown disturbance D of the multi-DOF manipulator (2)–(3) is assumed to be bounded, i.e., $\dot{D} \le \bar{d}$ with $\bar{d} > 0$.

Remark 1: In the dynamical model of the multi-DOF manipulator (2)–(3), it is inevitable that there exist deviations between the modeling parameters and real-world implementation, which is the uncertainty. The accuracy of the proposed fixed-time disturbance observer will decrease if the deviations between modeling parameters and real-world implementation are large.

Note that we minimize the deviations between modeling parameters and real-world implementation by providing sufficiently accurate modeling parameters of the multi-DOF manipulator. Moreover, the uncertainty of model is considered as a part of the unknown disturbance and is estimated and compensated by the proposed fixed-time disturbance observer and the fixed-time error feedback controller, respectively.

III. Main Results

In this section, the fixed-time angle tracking control strategy is developed to achieve the desired angle tracking control performances of the multi-DOF manipulator driven by PAMs. To this end, the fixed-time disturbance observer is designed to estimate the unknown disturbance of the multi-DOF manipulator (2)–(3), and the fixed-time controller is proposed to compensate the unknown disturbance of the multi-DOF manipulator (2)–(3).

A. Fixed-Time Disturbance Observer

According to the multi-DOF manipulator (2)–(3), the fixed-time disturbance observer is designed as

$$\dot{Z} = \hat{D} + \mathcal{B}_0 U + \lambda_1 \left(\left\lceil \frac{\sigma_1}{p_1} \right\rfloor + \left\lceil \frac{\sigma_1}{p_1} \right\rfloor^{\mu} + \left\lceil \frac{\sigma_1}{p_1} \right\rfloor^{2-\mu} \right) \tag{5}$$

$$\dot{\hat{D}} = \frac{\lambda_2}{p_1} \left(\left\lceil \frac{\sigma_1}{p_1} \right\rfloor + \left\lceil \frac{\sigma_1}{p_1} \right\rfloor^{2\mu-1} + \left\lceil \frac{\sigma_1}{p_1} \right\rfloor^{3-2\mu} \right) \tag{6}$$

where Z is the estimate of X_2, $\hat{D}$ is the estimate of D, $\sigma_1 = X_2 - Z$, $p_1 > 0$, $\lambda_1 > 0$, $\lambda_2 > 0$ and $(1/2) < \mu < 1$ are the adjustable parameters.

Define $\sigma_2 = D - \hat{D}$, $\varepsilon_j = (\sigma_j(p_1 t)/p_1^{2-j})$, $j = 1, 2$. According to the multi-DOF manipulator (2)–(3) and the fixed-time disturbance observer (5)–(6), the estimation error system is expressed as

$$\dot{\varepsilon}_1 = \varepsilon_2 - \lambda_1(\lceil \varepsilon_1 \rfloor + \lceil \varepsilon_1 \rfloor^{\mu} + \lceil \varepsilon_1 \rfloor^{2-\mu}) \tag{7}$$

$$\dot{\varepsilon}_2 = p_1 \dot{D} - \lambda_2(\lceil \varepsilon_1 \rfloor + \lceil \varepsilon_1 \rfloor^{2\mu-1} + \lceil \varepsilon_1 \rfloor^{3-2\mu}). \tag{8}$$

Theorem 1: Consider the multi-DOF manipulator (2)–(3) and the fixed-time disturbance observer (5)–(6), for the given adjustable parameters $p_1 > 0$, $\lambda_1 > 0$, $\lambda_2 > 0$ and $(1/2) < \mu < 1$, then the estimation error system (7)–(8) is fixed-time stable.

Proof: Denote the estimation error vector as $\varepsilon = [\varepsilon_1, \varepsilon_2]^T$, $\dot{\varepsilon} = \Phi(\varepsilon, \Delta)$, where $\Delta = p_1 \dot{D}$. The proof process consists of two steps. Step 1: When the unknown disturbance D is a constant, i.e., $\Delta = 0$, we will show the nominal estimation error system $\dot{\varepsilon} = \Phi(\varepsilon, 0)$ is fixed-time stable. Step 2: When the unknown disturbance D vary quickly, i.e., $\Delta \neq 0$, then the perturbed system $\dot{\varepsilon} = \Phi(\varepsilon, \Delta)$ is fixed-time convergent.

Step 1: Consider the nominal estimation error system $\dot{\varepsilon} = \Phi(\varepsilon, 0)$. Define vectors

$$\Phi_0(\varepsilon, 0) = \left[-\lambda_1 \lceil \varepsilon_1 \rfloor^{\mu} + \varepsilon_2, -\lambda_2 \lceil \varepsilon_1 \rfloor^{2\mu-1} \right]^T$$
$$\Phi_\infty(\varepsilon, 0) = \left[-\lambda_1 \lceil \varepsilon_1 \rfloor^{2-\mu} + \varepsilon_2, -\lambda_2 \lceil \varepsilon_1 \rfloor^{3-2\mu} \right]^T.$$

Since $(1/2) < \mu < 1$, then the inequalities $0 < 2\mu - 1 < 1 < 2 - \mu$ and $0 < 2\mu - 1 < 1 < 3 - 2\mu$ hold. Therefore, $\Phi_0(\varepsilon, 0)$ and $\Phi_\infty(\varepsilon, 0)$ can be considered as the approximating functions of $\Phi(\varepsilon, 0)$ in 0-limit and ∞-limit, respectively. Moreover, it can be found that $\Phi_0(\varepsilon, 0)$ and $\Phi_\infty(\varepsilon, 0)$ are homogeneous with degree $\lambda_0 = \mu - 1$ and $\lambda_\infty = 1 - \mu$ with respect to the weight vector $d_0 = [1, \mu]^T$ and $d_\infty = [1, 2 - \mu]^T$, respectively. Thus, we can conclude that $\Phi(\varepsilon, 0)$ is homogenous in bi-limit with triples $(d_0, \lambda_0, \Phi_0(\varepsilon, 0))$ and $(d_\infty, \lambda_\infty, \Phi_\infty(\varepsilon, 0))$.

Select the following Lyapunov function

$$V_d = \lambda_2(2-\mu)|\varepsilon_1|^{2\mu} + \lambda_2 \mu |\varepsilon_1|^{4-2\mu} + \lambda_2 \mu (2-\mu)|\varepsilon_1|^2 + \mu(2-\mu)|\varepsilon_2|^2$$

and it is easy to find that V_d is positive definite, continuous differentiable and radially unbounded for any $(1/2) < \mu < 1$. The derivative of V_d is expressed as

$$\begin{aligned} \dot{V}_d &= \tilde{\mu}(\lambda_2 \lceil \varepsilon_1 \rfloor^{2\mu-1} + \lambda_2 \lceil \varepsilon_1 \rfloor^{3-2\mu} + \lambda_2 \lceil \varepsilon_1 \rfloor)\dot{\varepsilon}_1 + \tilde{\mu} \lceil \varepsilon_2 \rfloor \dot{\varepsilon}_2 \\ &= -\tilde{\mu}\lambda_1\lambda_2 \left[|\varepsilon_1|^{3\mu-1} + |\varepsilon_1|^{2\mu} + |\varepsilon_1|^{\mu+1} \right] \\ &\quad - \tilde{\mu}\lambda_1\lambda_2 \left[|\varepsilon_1|^{3-\mu} + |\varepsilon_1|^{4-2\mu} + |\varepsilon_1|^{5-3\mu} \right] \\ &\quad - \tilde{\mu}\lambda_1 \left[|\varepsilon_1|^{2\mu} + |\varepsilon_1|^2 + |\varepsilon_1|^{3-\mu} \right] \end{aligned}$$

where $\tilde{\mu} = 2\mu(2-\mu)$. Since $\lambda_1 > 0$, $\lambda_2 > 0$ and $(1/2) < \mu < 1$, $\dot{V}_d < 0$ is satisfied, and the only invariant set of the nominal estimation error system $\dot{\varepsilon} = \Phi(\varepsilon, 0)$ is $\varepsilon_1 = \varepsilon_2 = 0$. According to LaSalle invariance principle, we can conclude that the nominal estimation error system $\dot{\varepsilon} = \Phi(\varepsilon, 0)$ is globally asymptotically stable.

For the nominal estimation error system $\dot{\varepsilon} = \Phi_0(\varepsilon, 0)$, select the following Lyapunov function

$$V_{d0} = \lambda_2 \mu |\varepsilon_1|^{2\mu} + \mu^2 |\varepsilon_2|^2.$$

The derivative of V_{d0} is expressed as

$$\begin{aligned} \dot{V}_{d0} &= 2\mu^2 \lambda_2 |\varepsilon_1|^{2\mu-1} \dot{\varepsilon}_1 + 2\mu^2 |\varepsilon_2| \dot{\varepsilon}_2 \\ &= -2\lambda_1\lambda_2\mu^2 |\varepsilon_1|^{3\mu-1}. \end{aligned}$$

Obviously, $\dot{V}_{d0} < 0$ is satisfied except on $\varepsilon_1 = 0$. According to LaSalle invariance principle, the nominal estimation error system $\dot{\varepsilon} = \Phi_0(\varepsilon, 0)$ is globally asymptotically stable. Similarly, we can also conclude that the nominal estimation error system $\dot{\varepsilon} = \Phi_\infty(\varepsilon, 0)$ is globally asymptotically stable.

Since the nominal estimation error systems $\dot{\varepsilon} = \Phi_0(\varepsilon, 0)$ and $\dot{\varepsilon} = \Phi_\infty(\varepsilon, 0)$ are globally asymptotically stable, $\lambda_0 < 0 < \lambda_\infty$ holds with $(1/2) < \mu < 1$. Therefore, the nominal estimation error system $\dot{\varepsilon} = \Phi(\varepsilon, 0)$ is fixed-time stable according to Lemma 1. Moreover, there exists a continuous, positive, and proper function $V(\varepsilon)$, such that $\partial V / \partial \varepsilon_j$ is homogeneous in bi-limit with triples $(d_0, \lambda_{V_0} - d_{0i}, (\partial V_0 / \partial \varepsilon_j))$ and $(d_\infty, \lambda_{V_\infty} - d_{\infty i}, \partial V_\infty / \partial \varepsilon_j)$, and $(\partial V / \partial \varepsilon_j)\Phi(\varepsilon, 0) < 0$, where $\lambda_{V_0} > \max\{1, \mu\}$, $\lambda_{V_\infty} > \max\{1, 2 - \mu\}$, $j = 1, 2$.

Step 2: Consider the perturbed system $\dot{\varepsilon} = \Phi(\varepsilon, \Delta)$ which is homogeneous in bi-limit with triples $((d_0, \delta_0), \lambda_0, \Phi_0(\varepsilon, \Delta))$ and $((d_\infty, \delta_\infty), \lambda_\infty, \Phi_\infty(\varepsilon, \Delta))$, respectively, where $\delta_0 = 2\mu - 1$, $\delta_\infty = 3 - 2\mu$. According to Lemma 2, the following inequality is obtained

$$\frac{\partial V}{\partial \varepsilon}\Phi(\varepsilon,\Delta) \le -a_V \Upsilon\left(V^{\frac{\lambda_{V_0}+\lambda_0}{\lambda_{V_0}}}, V^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\lambda_{V_\infty}}}\right) + a_\Delta \Upsilon\left(|\Delta|^{\frac{\lambda_{V_0}+\lambda_0}{\delta_0}}, |\Delta|^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\delta_\infty}}\right)$$

where $a_V > 0$, $a_\Delta > 0$. Since the derivative of the unknown disturbance D is bounded, the term $\Delta = |p_1 \dot{D}| < 1$ can be satisfied by selecting appropriate p_1, then the following inequality is obtained

$$\begin{aligned}\Upsilon\left(|\Delta|^{\frac{\lambda_{V_0}+\lambda_0}{\delta_0}}, |\Delta|^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\delta_\infty}}\right) &= \Upsilon\left(|\Delta|^{\frac{3\mu-1}{2\mu-1}}, |\Delta|^{\frac{5-3\mu}{3-2\mu}}\right)\\ &= \frac{|\Delta|^{\frac{3\mu-1}{2\mu-1}}}{1+|\Delta|^{\frac{3\mu-1}{2\mu-1}}}\left(1+|\Delta|^{\frac{5-3\mu}{3-2\mu}}\right)\\ &< 2|\Delta|^{\frac{3\mu-1}{2\mu-1}}.\end{aligned}$$

When $V > 1$, one has that

$$\begin{aligned}\Upsilon\left(V^{\frac{\lambda_{V_0}+\lambda_0}{\lambda_{V_0}}}, V^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\lambda_{V_\infty}}}\right) &= \frac{1}{V^{-\frac{3\mu-1}{2\mu}}+1}\left(1+V^{\frac{5-3\mu}{4-2\mu}}\right)\\ &\ge \frac{1}{2}\left(1+V^{\frac{5-3\mu}{4-2\mu}}\right).\end{aligned}$$

Since $(3\mu - 1/2\mu - 1) > 1$ and $|\Delta|$ is small by selecting appropriate p_1, the condition $a_V > 4a_\Delta|\Delta|^{(3\mu-1/2\mu-1)}$ can be satisfied, then the following inequality is obtained

$$\begin{aligned}\frac{\partial V}{\partial \varepsilon}\Phi(\varepsilon,\Delta) &\le -\frac{a_V}{2}\left(1+|V|^{\frac{5-3\mu}{4-2\mu}}\right) + 2a_\Delta|\Delta|^{\frac{3\mu-1}{2\mu-1}}\\ &\le -\frac{a_V}{2}V^{\frac{5-3\mu}{4-2\mu}}.\end{aligned}$$

Then the estimation error trajectory will converge to $V = 1$ in the time

$$T_{d1} \le \frac{8-4\mu}{a_V(1-\mu)}.$$

When $V \le 1$, one has that

$$\Upsilon\left(V^{\frac{\lambda_{V_0}+\lambda_0}{\lambda_{V_0}}}, V^{\frac{\lambda_{V_\infty}+\lambda_\infty}{\lambda_{V_\infty}}}\right) \ge \frac{1}{2}V^{\frac{3\mu-1}{2\mu}} + \frac{1}{2}V^{\frac{3\mu-1}{2\mu}+\frac{5-3\mu}{4-2\mu}}$$

which yields

$$\begin{aligned}\frac{\partial V}{\partial \varepsilon}\Phi(\varepsilon,\Delta) \le &-\frac{a_V}{2}V^{\frac{3\mu-1}{2\mu}} + 2a_\Delta|\Delta|^{\frac{3\mu-1}{2\mu-1}}\\ &-\frac{a_V}{2}V^{\frac{3\mu-1}{2\mu}+\frac{5-3\mu}{4-2\mu}}.\end{aligned} \tag{9}$$

Define $\zeta = (4a_\Delta|\Delta|^{(3\mu-1/2\mu-1)}/a_V)^{(1/3\mu-1/2\mu+5-3\mu/4-2\mu)}$. If $V > \zeta$, then the inequality (9) is simplified as

$$\frac{\partial V}{\partial \varepsilon}\Phi(\varepsilon,\Delta) \le -\frac{a_V}{2}V^{\frac{3\mu-1}{2\mu}}.$$

The error trajectory will converge to $V = \zeta$ in the time

$$T_{d2} \le \frac{4\mu}{a_V(1-\mu)}.$$

Based on above discussions, the estimation error vector ε will converge the bounded region ζ in the time

$$T_d = T_{d1} + T_{d2} \le \frac{8}{a_V(1-\mu)} \tag{10}$$

which does not depend on initial conditions. Therefore, the estimation error system (7)–(8) is fixed-time stable. This completes the proof.

B. Fixed-Time Controller

Define the angle tracking error and the angular velocity tracking error as $\eta = X_1 - \theta_r$ and $\vartheta = X_2 - \dot{\theta}_r$, respectively, where θ_r is the desired tracking angle. The angular velocity tracking error system is expressed as

$$\dot{\vartheta} = \mathcal{B}_0 U + D - \ddot{\theta}_r. \tag{11}$$

The fixed-time controller is designed as

$$U = \frac{1}{\mathcal{B}_0}\left[-\lambda_3\left(\left\lceil\frac{\vartheta}{p_2}\right\rfloor^{1+\kappa} + \left\lceil\frac{\vartheta}{p_2}\right\rfloor^{1-\kappa}\right) - \lambda_4\frac{\dot{\vartheta}}{p_2} - \hat{D} + \ddot{\theta}_r\right] \tag{12}$$

where $0 < p_2 < 1$, $0 < \kappa < 1$, $\lambda_3 > 0$ and $\lambda_4 > 0$ are the adjustable parameters.

Theorem 2: Consider the multi-DOF manipulator (2)–(3) under the fixed-time controller (12) and the fixed-time disturbance observer (5)–(6), if the parameters that satisfy $\lambda_3 > p_1^{1-\kappa}\rho$, then the angular velocity tracking error system (11) is fixed-time stable, where $\rho > 0$.

Proof: Choose the Lyapunov function as $V_c = \vartheta^2$. The derivative of V_c is expressed as

$$\dot{V}_c = -2\lambda_3 p_2\left|\frac{\vartheta}{p_2}\right|^{2-\kappa} - 2\lambda_3 p_2\left|\frac{\vartheta}{p_2}\right|^{2+\kappa} - \frac{2\lambda_4}{p_2}\vartheta\dot{\vartheta} - 2\vartheta D_f$$

where $D_f = \hat{D} - D$. Since the estimation error of the fixed-time disturbance observer (5)–(6) is bounded, then D_f is also bounded, i.e., $|D_f| < \tilde{\rho}$ when $t \le T_d$. When $t > T_d$, the estimation error $|D_f|$ will converge to ρ, i.e., $|D_f| \le \rho$. The derivative of V_c is calculated as

$$\dot{V}_c = -\frac{2\lambda_3 p_2}{1+\frac{2\lambda_4}{p_2}}\left|\frac{\vartheta}{p_2}\right|^{2-\kappa} - \frac{2\lambda_3 p_2}{1+\frac{2\lambda_4}{p_2}}\left|\frac{\vartheta}{p_2}\right|^{2+\kappa} - \frac{2}{1+\frac{2\lambda_4}{p_2}}\vartheta D_f.$$

When $0 < t < T_d$, the derivative of V_c is bounded by

$$\dot{V}_c \le -\frac{4\lambda_3 p_2}{1+\frac{2\lambda_4}{p_2}}\left|\frac{\vartheta}{p_2}\right|^2 - \frac{2}{1+\frac{2\lambda_4}{p_2}}\vartheta\tilde{\rho}. \tag{13}$$

According to inequality (13), $\dot{V}_c < 0$ holds if $|\vartheta| < (p_2\tilde{\rho}/2\lambda_3)$. Therefore, the angular velocity tracking error ϑ will not escape to infinity when $0 < t < T_d$. When $t > T_d$, the derivative of V_c is bounded by

$$\begin{aligned}\dot{V}_c \le &-\frac{2\lambda_3 p_2^{\kappa}}{2\lambda_4+p_2}V_c^{1-\frac{\kappa}{2}} - \frac{2\lambda_3 p_2^{-\kappa}}{2\lambda_4+p_2}V_c^{1+\frac{\kappa}{2}}\\ &+\frac{2p_2}{2\lambda_4+p_2}\rho V_c^{\frac{1}{2}}.\end{aligned} \tag{14}$$

TABLE II
ADJUSTABLE PARAMETERS OF THE FIXED-TIME DISTURBANCE OBSERVER (5)–(6) AND THE FIXED-TIME CONTROLLER (12)

Parameter	Value	Parameter	Value
λ_1	78	λ_2	152
μ_1	0.72	p_1	0.05
λ_3	115	λ_4	34
κ	0.65	p_2	0.05

For $V_c \geq 1$, the inequality (14) is recalculated as

$$\begin{aligned}\dot{V}_c \leq& -\frac{2p_2^{\kappa}}{2\lambda_4+p_2}V_c^{1-\frac{\kappa}{2}}\left(\lambda_3 - p_2^{1-\kappa}\rho V_c^{\frac{\kappa-1}{2}}\right)\\ &-\frac{2\lambda_3 p_2^{-\kappa}}{2\lambda_4+p_2}V_c^{1+\frac{\kappa}{2}}\\ \leq& -\frac{2\lambda_3 p_2^{-\kappa}}{2\lambda_4+p_2}V_c^{1+\frac{\kappa}{2}}.\end{aligned}$$

The angle tracking error trajectory will converge to $V_c = 1$ in the time $T_{c1} \leq (2\lambda_4 + p_2/\kappa\lambda_3 p_2^{-\kappa})$. For $V_c \leq 1$, if $V_c < (p_2^{1+\kappa}\rho/\lambda_3)^{(2/1+\kappa)} = \gamma_\theta$ then the inequality (14) is rewritten as

$$\begin{aligned}\dot{V}_c \leq& -\frac{2p_2^{-\kappa}}{2\lambda_4+p_2}V_c^{1+\frac{\kappa}{2}}\left(\lambda_3 - p_2^{1+\kappa}\rho V_c^{-\frac{\kappa+1}{2}}\right)\\ &-\frac{2\lambda_3 p_2^{\kappa}}{2\lambda_4+p_2}V_c^{1-\frac{\kappa}{2}}\\ \leq& -\frac{2\lambda_3 p_2^{\kappa}}{2\lambda_4+p_2}V_c^{1-\frac{\kappa}{2}}.\end{aligned}$$

The angle tracking error trajectory will converge to $V_c = \gamma_\theta$ in the time $T_{c2} \leq (2\lambda_4 + p_2/\kappa\lambda_3 p_2^{\kappa})$.

Based on above discussion, the angle tracking error trajectory ϑ will converge the bounded region γ_θ in the time

$$\begin{aligned}T_c \leq& T_d + T_{c1} + T_{c2}\\ \leq& \frac{8}{a_V(1-\mu)} + \frac{2\lambda_4+p_2}{\kappa\lambda_3 p_2^{-\kappa}} + \frac{2\lambda_4+p_2}{\kappa\lambda_3 p_2^{\kappa}}.\end{aligned}$$

Therefore, the angular velocity tracking error system (11) is fixed-time stable. This completes the proof.

IV. EXPERIMENT RESULTS

In this section, angle tracking experiments are carried out to evaluate the effectiveness of the fixed-time angle tracking control strategy proposed in this article. The adjustable parameters of the fixed-time disturbance observer (5)–(6) and the fixed-time controller (12) are shown in Table II. The supplying pressure of thirteen electric proportional valves is 0.6 MPa, and the sampling time is set as 0.01 s.

The selection process of the adjustable parameters in Table I is determined as follows. 1) Initialize the adjustable parameters, set all the adjustable parameters as 0. 2) By using the trial-and-error method, the adjustable parameters of the fixed-time disturbance observer (5)–(6) and the fixed-time controller (12) are tuned to minimize the estimation errors and the angle tracking errors, respectively. 3) Tune all the adjustable parameters appropriately until the small enough the errors are obtained. 4) To avoid contingencies in the experiment, it is necessary to carry out multiple experimental verification to further optimize the adjustable parameters. Moreover, the tradeoff between the convergence time T_c and the lifetime of the multi-DOF manipulator control system needs to be considered. Because the multi-DOF manipulator will receive a oscillation if the convergence time T_c is set small, which will reduce the lifetime of the multi-DOF manipulator control system. Note that to ensure the fairness of the comparative experimental results, the implementation of the comparative experiment are executed as follows. 1) Ensure the same experimental environment. The fixed-time angle tracking control method proposed in this article and the comparative methods are experimented by using the same multi-DOF manipulator control system. 2) Determine the selection range of the adjustable gains. The selection of the adjustable gains is based on the theoretical analysis results of the fixed-time angle tracking control method proposed in this article and the comparative methods. 3) Select the appropriate gain tuning method. By using different gain tuning methods, such as trial-and-error method, until the optimal experimental results are obtained. 4) Conduct multiple experiments. To avoid the randomness of experimental results, several experiments are carried out. The optimal experimental results obtained by the fixed-time angle tracking control method proposed in this article and the comparative methods are presented.

To verify the disturbance rejection performance of the fixed-time controller (12), a fixed-time controller without disturbance compensation is presented as

$$U = \frac{1}{\mathcal{B}_0}\left[-\lambda_3\left(\left\lceil\frac{\vartheta}{p_2}\right\rfloor^{1+\kappa} + \left\lceil\frac{\vartheta}{p_2}\right\rfloor^{1-\kappa}\right) - \lambda_4\frac{\dot{\vartheta}}{p_2} + \ddot{\theta}_r\right]. \tag{15}$$

To ensure the fairness of the experimental results, the adjustable parameters of the fixed-time controller without disturbance compensation are used in Table II. In the experiments, three sinusoidal signals $10^\circ \sin(0.1\pi t)$ are given as the desired tracking angle θ_r for Joint 1, Joint 2, and Joint 3, respectively. The comparison of angle tracking experimental results is shown in Fig. 4. Fig. 4(a), 4(c), and 4(e) show that the output angle X_1 obtained by using the fixed-time controller (12) more closely coincides with the desired tracking angle θ_r than the fixed-time controller (15) for Joint 1, Joint 2 and Joint 3, respectively. Fig. 4(b), 4(d), and 4(f) show that the angle tracking error η obtained by using the fixed-time angle tracking control strategy proposed in this article is smaller than the fixed-time controller (15) for Joint 1, Joint 2, and Joint 3, respectively. Therefore, we can conclude that the fixed-time angle tracking control strategy proposed in this article has good disturbance rejection performance.

In the first set of experiments, three sinusoidal signals $10^\circ \sin(0.2\pi t)$ are given as the desired tracking angle θ_r for

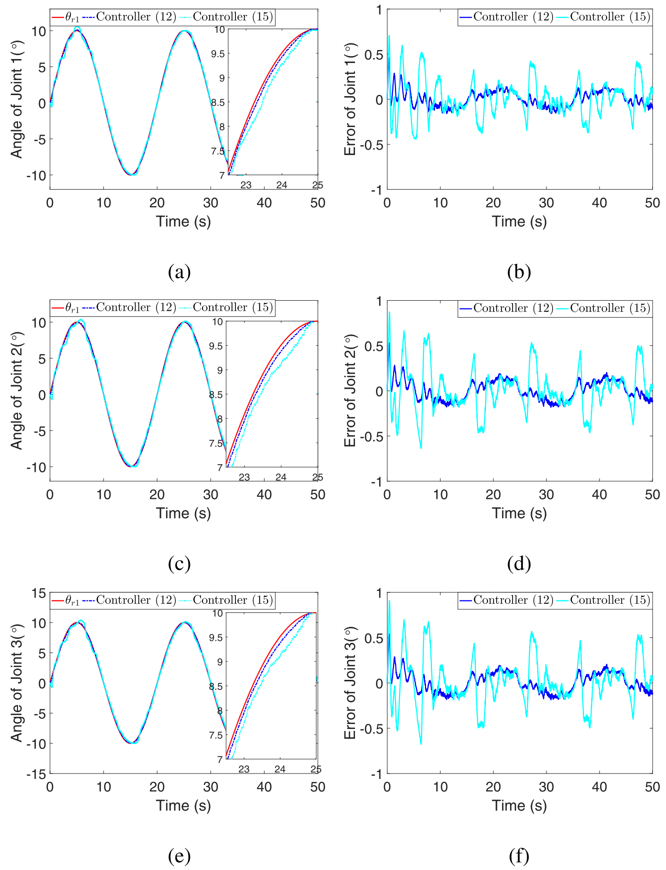


Fig. 4. Comparison of angle tracking experimental results. (a) Angle tracking of Joint 1. (b) Angle tracking error of Joint 1. (c) Angle tracking of Joint 2. (d) Angle tracking error of Joint 2. (e) Angle tracking of Joint 3. (f) Angle tracking error of Joint 3.

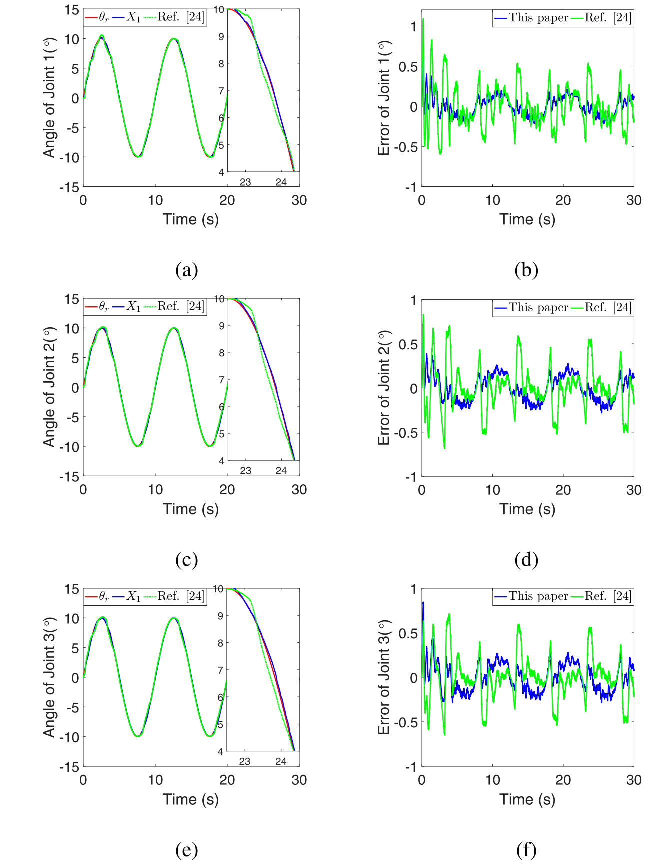


Fig. 5. First angle tracking experimental results. (a) Angle tracking of Joint 1. (b) Angle tracking error of Joint 1. (c) Angle tracking of Joint 2. (d) Angle tracking error of Joint 2. (e) Angle tracking of Joint 3. (f) Angle tracking error of Joint 3.

Joint 1, Joint 2, and Joint 3, respectively. The comparison of angle tracking experimental results obtained by using the fixed-time angle tracking control strategy proposed in this article and the one in [24] is shown in Fig. 5. Fig. 5(a), 5(c), and 5(e) show that the output angle X_1 obtained by using the fixed-time angle tracking control strategy proposed in this article more closely coincides with the desired tracking angle θ_r than the one in [24] for Joint 1, Joint 2 and Joint 3, respectively. Fig. 5(b), 5(d), and 5(f) show that the angle tracking error η obtained by using the fixed-time angle tracking control strategy proposed in this article is smaller than the one in [24] for Joint 1, Joint 2, and Joint 3, respectively.

In the second set of experiments, three sinusoidal signals $5^\circ \sin(0.4\pi t)$, $10^\circ \sin(0.4\pi t)$ and $15^\circ \sin(0.4\pi t)$ are given as the desired tracking angle θ_r for Joint 1, Joint 2, and Joint 3, respectively. The comparison of angle tracking experimental results obtained by using the fixed-time angle tracking control strategy proposed in this article and the one in [24] is shown in Fig. 6. Fig. 6(a), 6(c), and 6(e) show that the output angle X_1 obtained by using the fixed-time angle tracking control strategy proposed in this article more closely coincides with the desired tracking angle θ_r than the one in [24] for Joint 1, Joint 2, and Joint 3, respectively. Fig. 6(b), 6(d), and 6(f) show that the angle tracking error η obtained by using the fixed-time angle tracking control strategy proposed in this article is smaller than the one in [24] for Joint 1, Joint 2, and Joint 3, respectively.

In order to describe the angle tracking experimental results quantitatively, the root mean square error (RMSE) is given as

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{k=1}^{m} \eta_n^2}{m}} \tag{16}$$

where m is the total number of samples, η_n represents the angle tracking error under the nth sample. According to equality (16), the root mean square errors (RMSEs) of the first and second angle tracking experimental results are shown in Table III. The RMSEs of the first set of angle tracking experimental results are 0.4011, 0.4270, and 0.4302 obtained by using the fixed-time angle tracking control strategy proposed in this article for Joint 1, Joint 2, and Joint 3, respectively, the RMSEs are 0.7316, 0.7534, and 0.7866 obtained by using the strategy in [24] for Joint 1, Joint 2, and Joint 3, respectively. The RMSEs of the second set of angle tracking experimental results are 0.3966, 0.4113, and 0.4208 obtained by using the fixed-time angle tracking control strategy proposed in this article for Joint 1, Joint 2, and Joint 3, respectively, the RMSEs are 0.7413, 0.7701, and 0.8055 obtained by using the strategy in [24] for Joint 1, Joint 2, and Joint 3, respectively. It is clear that the RMSEs obtained by using the fixed-time angle tracking control

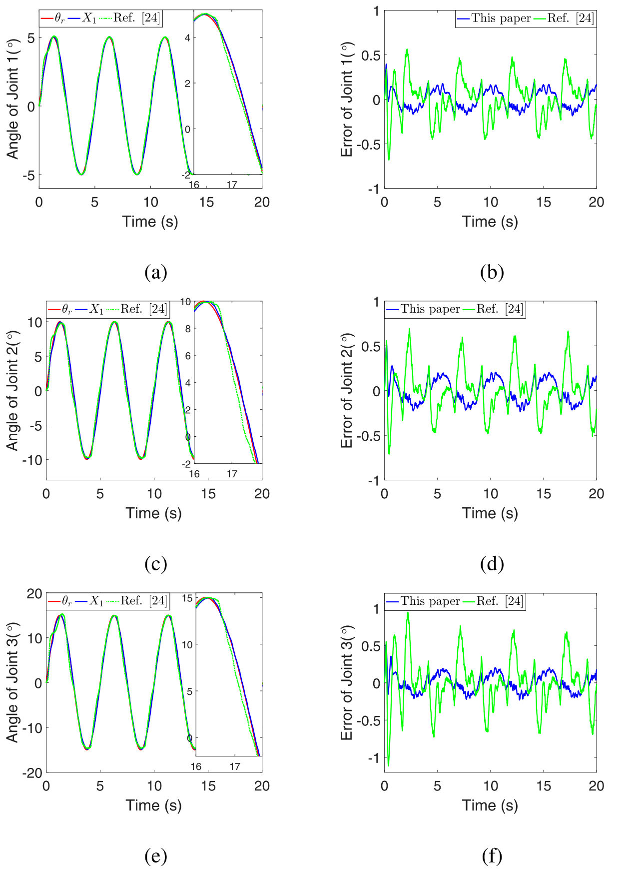


Fig. 6. Second angle tracking experimental results. (a) Angle tracking of Joint 1. (b) Angle tracking error of Joint 1. (c) Angle tracking of Joint 2. (d) Angle tracking error of Joint 2. (e) Angle tracking of Joint 3. (f) Angle tracking error of Joint 3.

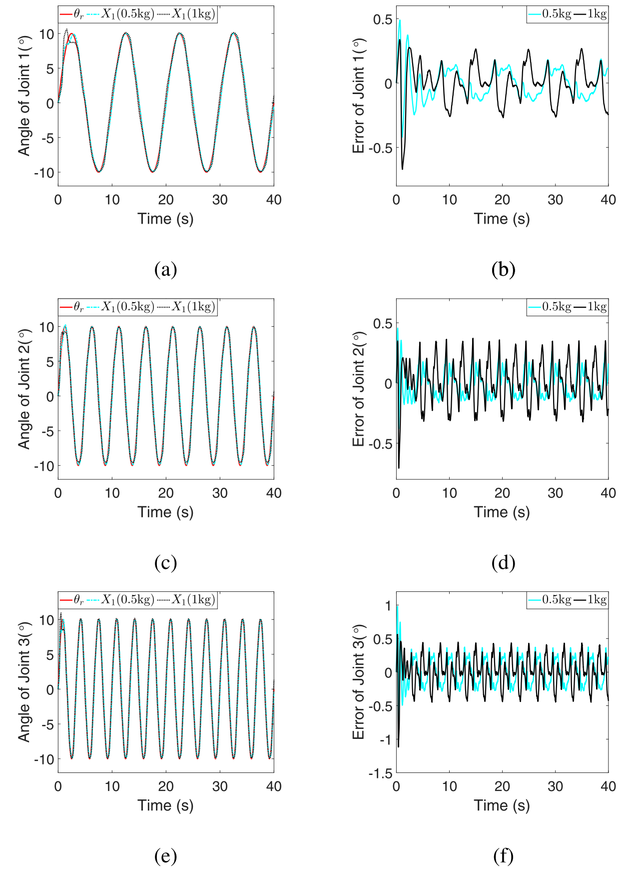


Fig. 7. Angle tracking experimental results with load. (a) Angle tracking of Joint 1. (b) Angle tracking error of Joint 1. (c) Angle tracking of Joint 2. (d) Angle tracking error of Joint 2. (e) Angle tracking of Joint 3. (f) Angle tracking error of Joint 3.

TABLE III
RMSEs OF THE FIRST AND SECOND ANGLE TRACKING EXPERIMENTAL RESULTS

Experiment	Strategy	Joint	RMSE (°)
First	This article	Joint 1	0.4011
		Joint 2	0.4270
		Joint 3	0.4302
	Ref. [24]	Joint 1	0.7316
		Joint 2	0.7534
		Joint 3	0.7866
Second	This article	Joint 1	0.3966
		Joint 2	0.4113
		Joint 3	0.4208
	Ref. [24]	Joint 1	0.7413
		Joint 2	0.7701
		Joint 3	0.8055

strategy proposed in this article are smaller than the one in [24], which implies that the fixed-time angle tracking control strategy proposed in this article is more effective.

To demonstrate the load characteristics of the fixed-time angle tracking control strategy proposed in this article, loads with 0.5 and 1 kg are grabbed by the pneumatic gripper, then three sinusoidal signals $10^\circ\ \sin(0.2\pi t)$, $10^\circ\ \sin(0.4\pi t)$, and $10^\circ\ \sin(0.6\pi t)$ are given as the desired tracking angle for Joint 1, Joint 2, and Joint 3, respectively. Angle tracking experimental results obtained by using the fixed-time angle tracking control strategy proposed in this article are shown in Fig. 7. Fig. 7(a), 7(c), and 7(e) show that the output angle X_1 obtained by using the fixed-time angle tracking control strategy proposed in this article closely coincides with the desired tracking angle $\theta_r(t)$ for Joint 1, Joint 2, and Joint 3 despite existence of the different loads, respectively. Fig. 7(b), 7(d), and 7(f) show that the angle tracking error η obtained by using the fixed-time angle tracking control strategy proposed in this article remains at 0.3°, 0.4°, and 0.5° for Joint 1, Joint 2, and Joint 3, respectively. It can be seen that the proposed fixed-time angle tracking control strategy proposed in this article is still effective under different loads.

To verify that the fixed-time angle tracking control strategy proposed in this article does not depend on the initial angle tracking error, the comparative experiment on convergence time is carried out. The comparative result of convergence time is shown in Fig. 8. Fig. 8 shows that the convergence time of the dynamic surface control strategy in [30] increases with the increase of the initial angle tracking error, however, the convergence time of the fixed-time angle tracking control strategy

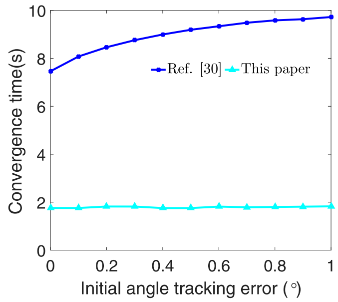


Fig. 8. Comparative result of convergence time.

proposed in this article is almost a constant as the initial angle tracking error increases.

Based on the angle tracking experimental results of different amplitudes, frequencies, loads and initial angle tracking errors, the fixed-time angle tracking control strategy proposed in this article is effective for achieving the desired angle tracking performances for the multi-DOF manipulator.

V. Conclusion

In this article, the angle tracking control problems for the multi-DOF manipulator driven by PAMs have been investigated. According to the dynamical model of the multi-DOF manipulator, a fixed-time disturbance observer has been proposed to estimate unknown disturbance, and a fixed-time controller is designed to track desired angle for the multi-DOF manipulator. Moreover, the fixed-time convergence of both the fixed-time disturbance observer and the multi-DOF manipulator with the fixed-time controller has been proved by using homogeneous theory and Lyapunov strategy, respectively. Finally, angle tracking experimental results of different amplitudes, frequencies, loads and initial angle tracking errors show that the desired angle tracking control performances have been achieved of the multi-DOF manipulator. In future works, we will consider applying the fixed-time angle tracking control strategy in the grasping process of the multi-DOF manipulator.

References

[1] R. Robinson, C. Kothera, and N. Wereley, “Variable recruitment testing of pneumatic artificial muscles for robotic manipulators,” *IEEE/ASME Trans. Mechatron.*, vol. 20, no. 4, pp. 1642–1652, Aug. 2015.

[2] Y. Cao, J. Huang, and C. Xiong, “Single-layer learning-based predictive control with echo state network for pneumatic-muscle-actuators-driven exoskeleton,” *IEEE Trans. Cogn. Devel. Syst.*, vol. 13, no. 1, pp. 80–90, Mar. 2021.

[3] J. Wu, J. Huang, Y. Wang, and K. Xing, “Energy-based motion control for pneumatic artificial muscle actuated robots with experiments,” *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7295–7306, Jul. 2022.

[4] Y. Yuan, Z. Wang, Y. Yu, L. Guo, and H. Yang, “Active disturbance rejection control for a pneumatic motion platform subject to actuator saturation: An extended state observer approach,” *Automatica*, vol. 107, pp. 353–361, Sep. 2019.

[5] Y. Liu, Y. Yang, Y. Peng, S. Zhong, N. Liu, and H. Pu, “A light soft manipulator with continuously controllable stiffness actuated by a thin Mckibben pneumatic artificial muscle,” *IEEE/ASME Trans. Mechatron.*, vol. 25, no. 4, pp. 1944–1952, Aug. 2020.

[6] D. Liang, N. Sun, Y. Wu, G. Liu, and Y. Fang, “Fuzzy-sliding mode control for humanoid arm robots actuated by pneumatic artificial muscles with unidirectional inputs, saturations, and dead zones,” *IEEE Trans. Ind. Inform.*, vol. 18, no. 5, pp. 3011–3021, May 2022.

[7] H. Choi, C. Lee, and Y. Baek, “Design and validation of a two-degree-of-freedom powered ankle-foot orthosis with two pneumatic artificial muscles,” *Mechatron.*, vol. 72, Dec. 2020, Art. no. 102469.

[8] M. Chang, “An adaptive self-organizing fuzzy sliding mode controller for a 2-DOF rehabilitation robot actuated by pneumatic muscle actuators,” *Control Eng. Practice*, vol. 18, no. 1, pp. 13–22, 2010.

[9] M. Zhang, J. Huang, Y. Cao, C. Xiong, and S. Mohammed, “Echo state network-enhanced super-twisting control of passive gait training exoskeleton driven by pneumatic muscles,” *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 6, pp. 5107–5118, Dec. 2022.

[10] D. Quy-Thinh, and T. Le, “Discrete-time sliding mode control based on exponential reaching law of a pneumatic artificial muscle actuator,” *J. Mech. Eng.*, vol. 19, no. 1, pp. 221–237, 2022.

[11] Y. Cao et al., “Adaptive proxy-based robust control integrated with nonlinear disturbance observer for pneumatic muscle actuators,” *IEEE/ASME Trans. Mechatron.*, vol. 25, no. 4, pp. 1756–1764, Aug. 2020.

[12] G. Liu, N. Sun, T. Yang, and Y. Fang, “Reinforcement learning-based prescribed performance motion control of pneumatic muscle actuated robotic arms with measurement noises,” *IEEE Trans. Syst., Man, Cybern.: Syst.*, vol. 53, no. 3, pp. 1801–1812, Mar. 2023.

[13] Z. Huang, and M. Chen, “Coordinated disturbance observer-based flight control of fixed-wing UAV,” *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 8, pp. 3545–3549, Aug. 2022.

[14] Y. Cui, J. Qiao, Y. Zhu, X. Yu, and L. Guo, “Velocity-tracking control based on refined disturbance observer for gimbal servo system with multiple disturbances,” *IEEE Trans. Ind. Electron.*, vol. 69, no. 10, pp. 10311–10321, Oct. 2022.

[15] K. Yu, and Z. Wang, “Improved deadbeat predictive current control of dual three-phase variable-flux PMSM drives with composite disturbance observer,” *IEEE Trans. Power Electron.*, vol. 37, no. 7, pp. 8310–8321, Jul. 2022.

[16] Z. Gao, Y. Zhang, and G. Guo, “Prescribed-time control of vehicular platoons based on a disturbance observer,” *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 9, pp. 3789–3793, Sep. 2022.

[17] Y. Wang, S. Xu, and C. Ahn, “Finite-time composite antidisturbance control for T–S fuzzy nonhomogeneous Markovian jump systems via asynchronous disturbance observer,” *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 11, pp. 5051–5057, Nov. 2022.

[18] Z. Peng, D. Wang, and J. Wang, “Data-driven adaptive disturbance observers for model-free trajectory tracking control of maritime autonomous surface ships,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5584–5594, Dec. 2021.

[19] H. Wang, K. Xu, and H. Zhang, “Adaptive finite-time tracking control of nonlinear systems with dynamics uncertainties,” *IEEE Trans. Autom. Control*, vol. 68, no. 9, pp. 5737–5744, Sep. 2023.

[20] Q. Yao, “Disturbance observer-based robust fixed-time integrated trajectory tracking control for space manipulator,” *Robotica*, vol. 40, no. 9, pp. 3214–3232, 2022.

[21] G. Guo, and P. Zhang, “Asymptotic stabilization of USVs with actuator dead-zones and yaw constraints based on fixed-time disturbance observer,” *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 302–316, Jan. 2020.

[22] C. Zhang, G. Zhang, and Q. Dong, “Fixed-time disturbance observer-based nearly optimal control for reusable launch vehicle with input constraints,” *ISA Trans.*, vol. 122, pp. 182–197, Mar. 2022.

[23] X. Ai, “Fixed-time consensus disturbance rejection for high-order nonlinear multi-agent systems with input saturation,” *Trans. Inst. Meas. Control*, vol. 44, no. 12, pp. 2443–2453, 2022.

[24] J. Zhang, F. Xu, X. Liu, S. Gu, and H. Geng, “Fixed-time dynamic surface control for pneumatic manipulator system with unknown disturbances,” *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 10890–10897, Oct. 2022.

[25] A. Polyakov, “Nonlinear feedback design for fixed-time stabilization of linear control systems,” *IEEE Trans. Autom. Control*, vol. 57, no. 8, pp. 2106–2110, Aug. 2012.

[26] A. Bacciotti, and L. Rosier, *Lyapunov Functions and Stability in Control Theory*, New York, NY, USA: Springer-Verlag, 2006.

[27] V. Andrie, L. Praly, and A. Astolfi, “Homogeneous approximation, recursive observer design, and output feedback,” *SIAM J. Control Optim.*, vol. 47, no. 4, pp. 1814–1850, 2008.

[28] B. Tian, Z. Zuo, X. Yan, and H. Wang, "A fixed-time output feedback control scheme for double integrator systems," *Automatica*, vol. 80, pp. 17–24, Jun. 2017.

[29] Y. Huang, M. Zhu, Z. Zheng, and M. Feroskhan, "Fixed-time autonomous shipboard landing control of a helicopter with external disturbances," *Aerosp. Sci. Technol.*, vol. 84, pp. 18–30, Jan. 2019.

[30] J. Wu, Huang, Y. Wang, and K. Xing, "Nonlinear disturbance observer-based dynamic surface control for trajectory tracking of pneumatic muscle system," *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 2, pp. 440–455, Mar. 2014.

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